

Name: Matthew Howman
ID Number: 1635699

Math 166
U-Substitution Gateway 12:30 pm

1. Evaluate the following integral, and clearly show all your work. Then, put your final answer in the box provided. You might lose credit if your answer is not clearly written in the provided box.

$$\int_2^e \frac{(\ln(x))^2}{x} dx$$

$$u = \ln x$$

$$\int_2^e u^2 du$$

$$du = \frac{1}{x} dx$$

$$\cancel{\frac{1}{x}} du = dx$$

$$\left[\frac{u^3}{3} \right]_2^e$$

$$\frac{e^3}{3} - \frac{2^3}{3} \rightarrow \frac{8}{3}$$

$$\frac{e^3}{3} - \frac{8}{3}$$

Final answer = $\frac{e^3}{3} - \frac{8}{3}$

Name: Drew Olson
 ID Number: 1630325

Math 166
 U-Substitution Gateway 12:30 pm

1. Evaluate the following integral, and clearly show all your work. Then, put your final answer in the box provided. You might lose credit if your answer is not clearly written in the provided box.

$$\int_2^e \frac{(\ln(x))^2}{x} dx$$

$$\frac{(\ln x)^2}{x} \quad \text{---} \quad \frac{u^2}{x} \quad du \quad \text{---}$$

$$u = \ln x$$

$$du = \frac{1}{x}$$

$$dx = du \cdot x$$

$$1 \int u^2$$

$$1 \int \frac{u^3}{3}$$

$$\frac{1}{3} u^3 + C$$

$$\frac{1}{3} \ln(x) + C$$

$$\ln(x^{1/3}) + C$$

Final answer = $\boxed{\frac{1}{3} \ln(2) - \frac{1}{3} \ln(e)}$

Name: Cyprian Brantuma
 ID Number: 1438960

Math 166
 U-Substitution Gateway 12:30 pm

1. Evaluate the following integral, and clearly show all your work. Then, put your final answer in the box provided. You might lose credit if your answer is not clearly written in the provided box.

$$\int_2^e \frac{(\ln(x))^2}{x} dx$$

$$u = \ln x$$

$$du = \frac{1}{x} dx$$

$$\int_{\ln 2}^1 u^2$$

$$\ln e = 1$$

$$\ln 2 = \ln 2$$

$$\frac{1}{3} u^3$$

$$\left[\frac{1}{3} (\ln x)^3 \right]_{\ln 2}^1$$

$$\frac{1}{3} \left((\ln(1))^3 - (\ln(\ln(2)))^3 \right)$$

Final answer = $\boxed{\frac{1}{3} \left((\ln(1))^3 - (\ln(\ln(2)))^3 \right)}$